Imagine a Burger Shop

Step 1: Single Worker (No scaling, no load balancing)

You have **1 worker** taking orders and making burgers.

- 5 customers → OK
- 50 customers → very slow, long lines

Problem: one person can't handle too much work.

Scaling (Adding more workers)

Now you **hire more workers**.

- 1 worker → 5 workers
- Each one can make burgers

This is **scaling** (specifically *horizontal scaling*)

Now the shop can handle more customers.

Load Balancing (Assigning customers properly)

But wait… customers are still randomly choosing lines.

- One worker gets 20 customers
- Another gets 2 customers

Still inefficient!

So you add a **manager at the entrance**:

- Directs each customer to the **shortest line**

That manager = **load balancer**

Put together

- **Scaling** = hiring more workers
- **Load balancing** = assigning customers evenly

Final idea

Concept	In Burger Shop	In Tech System
Scaling	Add more workers	Add more servers
Load Balancing	Manager directs customers	Distributes requests

Key idea

If you **only scale** (add workers) but don't balance:
→ some workers are overloaded

If you **only balance** but have few workers:
→ still slow

You need **both together**